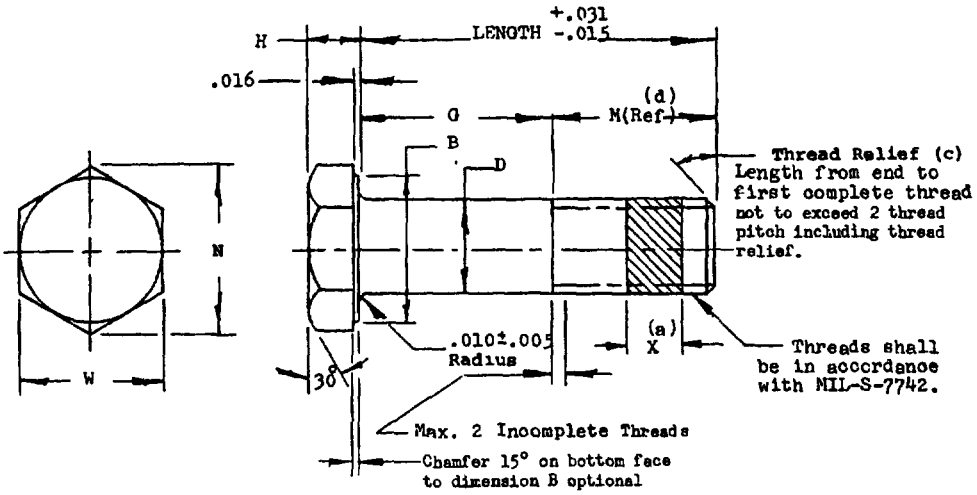
FED. SUP CLASS
5306



DIA DASH NO	(b) THREAD UNF 3A	D DIA		(f) B DIA	H		M Ref	N	W		DBL SHEAR STRENGTH POUNDS	(e) ULT TENSILE STRENGTH POUNDS
		MAX	MIN	MIN.	MAX	MIN		MIN	MAX	MIN	MIN	MIN
3	#10-32	189	186	.335	141	109	.469	416	.377	.365	4,250	2,420
4	1/4-28	249	.246	.398	172	.140	506	488	440	428	7,360	4,420
5	5/16-24	312	309	460	204	172	531	558	502	.490	11,500	7,100
6	3/8-24	374	371	.523	235	203	641	.630	.565	.553	16,560	10,790
7	7/16-20	437	433	.585	266	234	656	701	.627	.615	22,500	14,600
8	1/2-20	499	495	710	.297	.265	781	843	.752	.740	29,400	19,810
9	9/16-18	562	.558	835	328	.296	906	986	877	.865	37,400	25,060
10	5/8-18	624	.620	898	360	328	953	1 058	940	928	46,000	31,600
12	3/4-16	749	744	1.023	422	390	969	1.200	1 066	1.053	66,300	46,100
14	7/8-14	874	869	1 210	.485	.453	1.250	1 413	1 253	1 240	90,100	62,800
16	1-12	999	993	1 398	547	515	1 375	1.628	1 441	1.428	117,800	82,100
18	1-1/8-12	1 124	1 118	1 585	610	578	1 500	1 841	1 628	1 615	147,500	105,750
20	1-1/4-12	1 249	1 243	1 772	672	640	1 687	2 054	1.815	1.802	182,100	133,000

(a) Location of effective locking area X shall be in accordance with MS15981
(b) Threads shall be formed by any single rolling process and in accordance with Specification MIL-S-7742
(c) Chamfer 45° ±5° (See MIL-STD-8 for clarification of angular tolerance) Chamfer to extend to root of thread.
(d) Reference dimensions are for design purposes only and are not an inspection of manufacturing requirement.
(e) Tensile values based on 125 KSI ultimate tensile strength Stress area = $3.1416\left(\frac{E}{2} - \frac{3H}{16}\right)^2$. See Handbook H-28 (1957) Part 1
(f) The diameter of the washer face shall not exceed the actual width across flats

MATERIAL Alloy Steel - See Procurement Specification

HARDNESS· Rockwell C26-C32

PLATING Cadmium Plated in accordance with Specification QQ-P-416, Type II, Class 3

SURFACE ROUGHNESS The surface of the shank, fillet and bearing area shall not exceed 125 micro-inches See MIL-STD-10.

LOCKING ELEMENT In accordance with Specification MIL-F-18240. Locking element area must pass thru a ring gage equal to maximum major thread diameter + 003" freely or with finger pressure

DESIGN AND USAGE LIMITATIONS See MS15981

DIMENSIONS In inches

EXAMPLE OF PART NUMBER· MS21094 -4008 = 1/4"-28 bolt, .500" grip length

FOR DESIGN FEATURE PURPOSES THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID

P.A. Navy - WP Other Cust USAF - 11	TITLE BOLT, SELF-LOCKING, 250° F , STEEL, 75 KSI Psu, 125 KSI Ftu, HEX HEAD	MILITARY STANDARD MS21094(ASG)
PROCUREMENT SPECIFICATION MIL-B-6812 & MIL-F-18240	SUPERSEDES	SHEET 1 OF 2

APPROVED 2 Nov 65 REVISED

Review activities: Navy - WP USAF - 11

This military standard is approved by the Department of the Air Force and the Bureau of Naval Weapons and is mandatory for use by these activities. All other military activities are required to employ this standard where suitable.

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FED SUP CLASS 5306

LENGTH DASH NO. (a)	GRIP G ±.010	LENGTH +.031 -.015												
		10-32	1/4-28	5/16-24	3/8-24	7/16-20	1/2-20	9/16-18	5/8-18	3/4-16	7/8-14	1-12	1-1/8-12	1-1/4-12
003	.187	.593												
004	.250	.656	.719											
006	.375	.781	.844	.906	1.016									
008	.500	.906	.969	1.031	1.141	1.156	1.281							
010	.625	1.031	1.094	1.156	1.266	1.281	1.406	1.531	1.578					
012	.750	1.156	1.219	1.281	1.391	1.406	1.531	1.656	1.703	1.719				
014	.875	1.281	1.344	1.406	1.516	1.531	1.656	1.781	1.828	1.844	2.125			
016	1.000	1.406	1.469	1.531	1.641	1.656	1.781	1.906	1.953	1.969	2.250	2.375		
018	1.125	1.531	1.594	1.656	1.766	1.781	1.906	2.031	2.078	2.094	2.375	2.500	2.625	
020	1.250	1.656	1.719	1.781	1.891	1.906	2.031	2.156	2.203	2.219	2.500	2.625	2.750	2.937
022	1.375	1.781	1.844	1.906	2.016	2.031	2.156	2.281	2.328	2.344	2.625	2.750	2.875	3.062
024	1.500	1.906	1.969	2.031	2.141	2.156	2.281	2.406	2.453	2.469	2.750	2.875	3.000	3.187
026	1.625	2.031	2.094	2.156	2.266	2.281	2.406	2.531	2.578	2.594	2.875	3.000	3.125	3.312
028	1.750	2.156	2.219	2.281	2.391	2.406	2.531	2.656	2.703	2.719	3.000	3.125	3.250	3.437
030	1.875	2.281	2.344	2.406	2.516	2.531	2.656	2.781	2.828	2.844	3.125	3.250	3.375	3.562
032	2.000	2.406	2.469	2.531	2.641	2.656	2.781	2.906	2.953	2.969	3.250	3.375	3.500	3.687
034	2.125	2.531	2.594	2.656	2.766	2.781	2.906	3.031	3.078	3.094	3.375	3.500	3.625	3.812
036	2.250	2.656	2.719	2.781	2.891	2.906	3.031	3.156	3.203	3.219	3.500	3.625	3.750	3.937
038	2.375	2.781	2.844	2.906	3.016	3.031	3.156	3.281	3.328	3.344	3.625	3.750	3.875	4.062
040	2.500	2.906	2.969	3.031	3.141	3.156	3.281	3.406	3.453	3.469	3.750	3.875	4.000	4.187
042	2.625	3.031	3.094	3.156	3.266	3.281	3.406	3.531	3.578	3.594	3.875	4.000	4.125	4.312
044	2.750	3.156	3.219	3.281	3.391	3.406	3.531	3.656	3.703	3.719	4.000	4.125	4.250	4.437
046	2.875	3.281	3.344	3.406	3.516	3.531	3.656	3.781	3.828	3.844	4.125	4.250	4.375	4.562
048	3.000	3.406	3.469	3.531	3.641	3.656	3.781	3.906	3.953	3.969	4.250	4.375	4.500	4.687
050	3.125	3.531	3.594	3.656	3.766	3.781	3.906	4.031	4.078	4.094	4.375	4.500	4.625	4.812
052	3.250	3.656	3.719	3.781	3.891	3.906	4.031	4.156	4.203	4.219	4.500	4.625	4.750	4.937
054	3.375	3.781	3.844	3.906	4.016	4.031	4.156	4.281	4.328	4.344	4.625	4.750	4.875	5.062
056	3.500	3.906	3.969	4.031	4.141	4.156	4.281	4.406	4.453	4.469	4.750	4.875	5.000	5.187
058	3.625	4.031	4.094	4.156	4.266	4.281	4.406	4.531	4.578	4.594	4.875	5.000	5.125	5.312
060	3.750		4.219	4.281	4.391	4.406	4.531	4.656	4.703	4.719	5.000	5.125	5.250	5.437
062	3.875		4.344	4.406	4.516	4.531	4.656	4.781	4.828	4.844	5.125	5.250	5.375	5.562
064	4.000		4.469	4.531	4.641	4.656	4.781	4.906	4.953	4.969	5.250	5.375	5.500	5.687
066	4.125		4.594	4.656	4.766	4.781	4.906	5.031	5.078	5.094	5.375	5.500	5.625	5.812
068	4.250		4.719	4.781	4.891	4.906	5.031	5.156	5.203	5.219	5.500	5.625	5.750	5.937
070	4.375		4.844	4.906	5.016	5.031	5.156	5.281	5.328	5.344	5.625	5.750	5.875	6.062
072	4.500		4.969	5.031	5.141	5.156	5.281	5.406	5.453	5.469	5.750	5.875	6.000	6.187
074	4.625			5.156	5.266	5.281	5.406	5.531	5.578	5.594	5.875	6.000	6.125	6.312
076	4.750			5.281	5.391	5.406	5.531	5.656	5.703	5.719	6.000	6.125	6.250	6.437
078	4.875			5.406	5.516	5.531	5.656	5.781	5.828	5.844	6.125	6.250	6.375	6.562
080	5.000			5.531	5.641	5.656	5.781	5.906	5.953	5.969	6.250	6.375	6.500	6.687
082	5.125			5.656	5.766	5.781	5.906	6.031	6.078	6.094	6.375	6.500	6.625	6.812
084	5.250			5.781	5.891	5.906	6.031	6.156	6.203	6.219	6.500	6.625	6.750	6.937
086	5.375			5.906	6.016	6.031	6.156	6.281	6.328	6.344	6.625	6.750	6.875	7.062
088	5.500			6.031	6.141	6.156	6.281	6.406	6.453	6.469	6.750	6.875	7.000	7.187
090	5.625					6.281	6.406	6.531	6.578	6.594	6.875	7.000	7.125	7.312
092	5.750					6.406	6.531	6.656	6.703	6.719	7.000	7.125	7.250	7.437
094	5.875					6.531	6.656	6.781	6.828	6.844	7.125	7.250	7.375	7.562
096	6.000					6.656	6.781	6.906	6.953	6.969	7.250	7.375	7.500	7.687
098	6.125					6.781	6.906	7.031	7.078	7.094	7.375	7.500	7.625	7.812
100	6.250					6.906	7.031	7.156	7.203	7.219	7.500	7.625	7.750	7.937
102	6.375					7.031	7.156	7.281	7.328	7.344	7.625	7.750	7.875	8.062
104	6.500							7.406	7.453	7.469	7.750	7.875	8.000	8.187
106	6.625							7.531	7.578	7.594	7.875	8.000	8.125	8.312
108	6.750							7.656	7.703	7.719	8.000	8.125	8.250	8.437
110	6.875							7.781	7.828	7.844	8.125	8.250	8.375	8.562
112	7.000							7.906	7.953	7.969	8.250	8.375	8.500	8.687
114	7.125							8.031	8.078	8.094	8.375	8.500	8.625	8.812
116	7.250												8.750	8.937
118	7.375												8.875	9.062
120	7.500												9.000	9.187
122	7.625												9.125	9.312
124	7.750												9.250	9.437
126	7.875												9.375	9.562
128	8.000												9.500	9.687

(a) Unassigned dash numbers shall not be used except those dash numbers included by the following coding note Dash numbers indicate preferred grip lengths in .125 increments for lengths longer than 2.50 inches intermediate grip lengths between those lengths listed may be specified in .062 increments by intermediate dash numbers Length and grip of a bolt with an odd number for the dash number will be .062 greater than shown for the bolt with the preceding even dash number

APPROVED 2 Nov 65 REVISED

P.A Navy - WP Other Cust USAF -11	TITLE BOLT, SELF-LOCKING, 250° F., STEEL, 75 KSI Fsu, 125 KSI Ftu, HEX HEAD	MILITARY STANDARD MS21094(ASG)
PROCUREMENT SPECIFICATION MIL-B-6812&MIL-F-18240	SUPERSEDES	SHEET 2 OF

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